Quantile Regression
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Bryan S. Graham

University of California - Berkeley

Means

Tycho Brahe attempted to reduced measurement error in astronomical observations by *averaging* repeated measures.

Quetelet, inventor of *l'homme moyen* (the "average man"), introduced the idea of the mean (expected value) as a "representative value".

Our typical notation of regression considers how the average of Y changes with $X = x$.

Means can be misleading

"Bill Gates walks into a bar..."

A distinction between *the* average person and a "typical" person...

The "typical" person is "in the middle".

[we] "shall seek out the middle place amongst all the arrows" (Edward Wright, 1599).

(Population) median

Let $Y \sim F_Y$ (e.g., the annual earnings of a randomly sampled American household).

Let $\theta_{1/2}$ be the income level such that half of American households earn less and half earn more.

If you guess that a randomly sampled household's income is $\theta_{1/2}$, your guess will be too low half the time and too high half the time.

The chance of under- and over-prediction coincides.

(Population) median (continued)

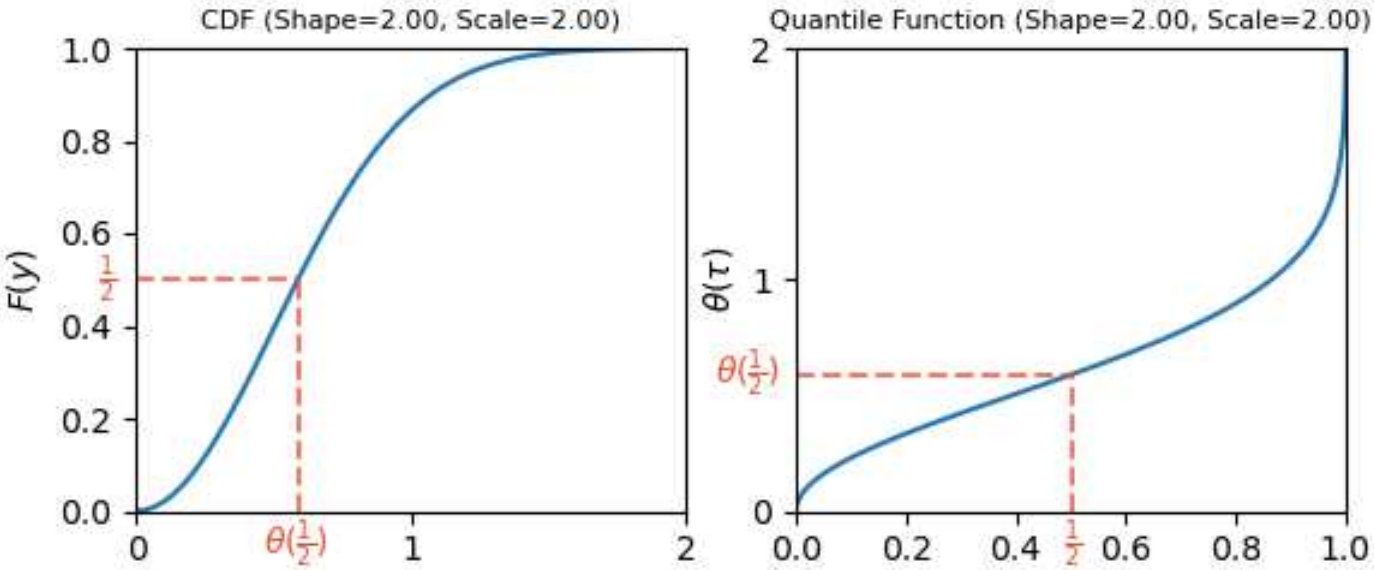
Let $F_Y(y) = \Pr(Y \leq y)$ denote the CDF.

Formally, $\theta_{1/2}$, the *population median*, solves:

$$\begin{aligned} \frac{1}{2} &= \Pr\left(Y \leq \theta_{1/2}\right) \\ &= F_Y\left(\theta_{1/2}\right). \end{aligned}$$

Assuming $F_Y(y)$ is strictly increasing we get

$$\theta_{1/2} = F_Y^{-1}\left(\frac{1}{2}\right).$$


CDF (Shape=2.00, Scale=2.00)
Quantile Function (Shape=2.00, Scale=2.00)
F(y)
θ(τ)
1.0
0.8
0.6
0.4
0.2
0.0
0
1
2
0.0
0.2
0.4
0.6
0.8
1.0
1/2
θ(1/2)

Quantiles

Let θ_τ for $\tau \in (0,1)$ denote the earnings level such that 100τ percent of households earn less that θ_τ and $100\,(1-\tau)$ percent of households earn more.

For example, $\theta_{0.10}$ gives the income level such that 10 percent of households earn less and 90 percent earn more.

$\theta_{3/4} - \theta_{1/4}$ equals the *inter-quartile range (IQR)*, a concept you may have encountered before.

Conditional quantiles

Let $F_{Y|X}(y|x)$ give the proportion of households who earn less than or equal to y *within the subpopulation where* $X = x$.

What fraction of households with college-educated heads ($X = x$) earn less that \$50,000 (y)?

$Q_{Y|X}(\tau|x)$ gives the income level such that, within households where $X = x$, 100τ percent of households earn less that $Q_{Y|X}(\tau|x)$ and $100(1-\tau)$ percent of households earn more.

By varying $X = x$ we trace out the τth *quantile regression* function.

Quantile regression

We typically assume that

$$Q_{Y|X}(\tau|\, x) = x'\beta_\tau.$$

Quantile regression provides a method for summarizing the entire conditional distribution of Y given X.

Can help us understand how changes in $X = x$ affect different parts of the distribution of Y.

Our goal is to outline a method for estimating β_τ and conducting inference.

Unconditional Quantiles: Estimation

Let $Y_1, Y_2, \ldots, Y_N$ be a random sample drawn from population F_Y.

Let $Y_{(1)}, Y_{(2)}, \ldots, Y_{(N)}$ be the associated order statistics; from smallest:

$$Y_{(1)} = \min \{Y_1, Y_2, \ldots, Y_N\},$$

to largest:

$$Y_{(N)} = \max \{Y_1, Y_2, \ldots, Y_N\}.$$

Unconditional Quantiles: Estimation

Let l be an integer such that

$$\frac{l}{N+1} \leq \tau < \frac{l+1}{N+1}.$$

We estimate θ_τ by the τth sample quantile:

$$\boxed{\widehat{\theta}_\tau = \frac{Y_{(l)} + Y_{(l+1)}}{2}.}$$

With $N = 10$ and $\tau = 1/2$ we find that $l = 5$ and

$$\widehat{\theta}_{1/2} = \frac{Y_{(5)} + Y_{(6)}}{2}.$$

Order statistics confidence interval

Let $j < k$ (so that $Y_{(j)} < Y_{(k)}$).

We can ask the question what is the *ex ante* chance that the (random interval) $\left[Y_{(j)}, Y_{(k)}\right]$ contains θ_τ?

What is the chance that the true median income in the United States lies between the 3rd and 7th order statistic in our sample of size $N = 10$?

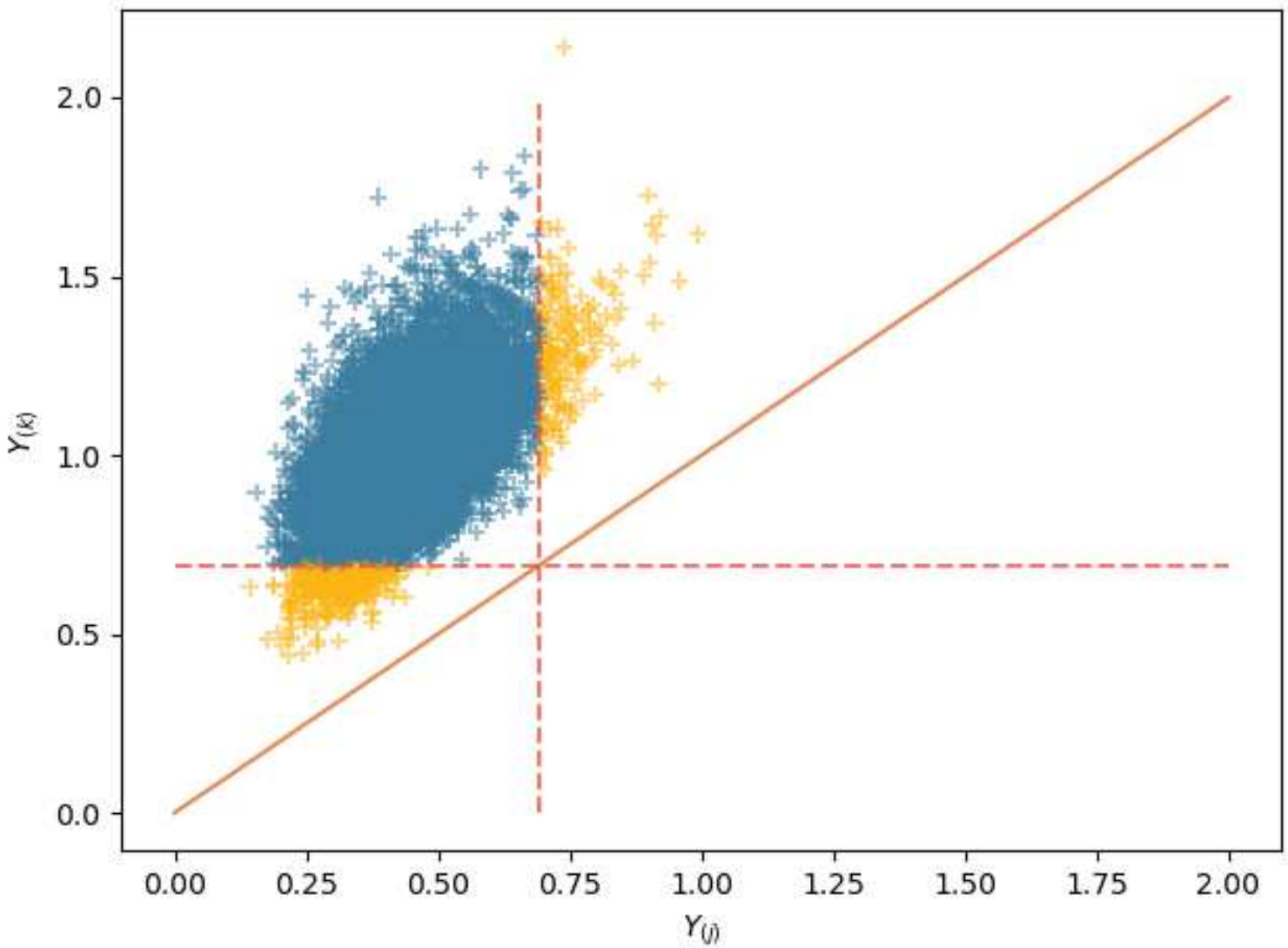
It turns out we can answer this question quite precisely!

Order statistics confidence interval

Observe that, since $Y_{(j)} < Y_{(k)}$ by construction, the $\left(Y_{(j)}, Y_{(k)}\right)$ pairs always "lie above" the 45-degree line.

A consequence of this observation is that:

$$\Pr\left(Y_{(j)} \leq \theta_\tau \leq Y_{(k)}\right) = \Pr\left(Y_{(j)} \leq \theta\left(\tau\right)\right) - \Pr\left(Y_{(k)} < \theta\left(\tau\right)\right).$$


2.0
1.5
1.0
0.5
0.0
$Y_{(k)}$
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
$Y_{(j)}$

Order statistics confidence interval

$$
\begin{aligned}
\Pr\left(Y_{(j)} \leq \theta_\tau\right) &= \Pr\left(j^{th} \text{ order statistic} \leq \theta_\tau\right) \\
&= \Pr\left(j \text{ or more observations} \leq \theta_\tau\right) \\
&= \sum_{i=j}^{N} \Pr\left(\text{exactly } i \text{ observations} \leq \theta_\tau\right) \\
&= \sum_{i=j}^{N} \binom{N}{i} F\left(\theta_\tau\right)^i \left[1 - F\left(\theta_\tau\right)\right]^{N-i} \\
&= \sum_{i=j}^{N} \binom{N}{i} \tau^i \left[1 - \tau\right]^{N-i}
\end{aligned}
$$

Order statistics confidence interval

Putting things together we get

$$\begin{aligned}\Pr\left(Y_{(j)} \leq \theta_\tau \leq Y_{(k)}\right) &= \sum_{i=j}^{N} \binom{N}{i} \tau^i \left[1-\tau\right]^{N-i} - \sum_{i=k}^{N} \binom{N}{i} \tau^i \left[1-\tau\right]^{N-i} \\ &= \sum_{i=j}^{k-1} \binom{N}{i} \tau^i \left[1-\tau\right]^{N-i}.\end{aligned}$$

We can use a table of the binomial distribution to evaluate the coverage of our confidence interval.

Order statistics confidence interval

With $N = 10$ and $\tau = 1/2$ we have

$$\begin{aligned}
\Pr\left(Y_{(2)} \leq \theta_{1/2} \leq Y_{(8)}\right) &= \sum_{i=2}^{8} \binom{N}{i} \left(\frac{1}{2}\right)^{i} \left[1 - \frac{1}{2}\right]^{10-i} \\
&= \sum_{i=2}^{8} \binom{N}{i} \left(\frac{1}{2}\right)^{10} \\
&= 0.9784.
\end{aligned}$$

The population median lines inside the interval $\left[Y_{(2)}, Y_{(8)}\right]$ in about 98 percent of (repeated) random samples of size 10.

Awesome!!!

Choosing j and k

When N is large it can be challenging to choose j and k such that coverage is approximately $1 - \alpha$.

Let $Z \sim \text{Binomial}(N, \tau)$.

Observe that the ex ante probability of observing between j and k successes across N independent Bernoulli trials, each with a success probability of τ equals

$$\Pr\left(j \leq Z \leq k\right) = \sum_{i=j}^{k-1} \binom{N}{i} \tau^i \left[1 - \tau\right]^{N-i}.$$

PLINKO!
2 DISKS $5
2 DISKS $5
Santa Cruz Beach Boardwalk
PLINKO RUL
REACH OVERHEAD AS HIGH A
ACCEPT FIRST PRIZE OR TRY
SECOND PRIZE IS THE ONE YO
STUCK DISK IS VOID AND MUS
TRIED AGAIN.
LARGE

Choosing j and k

Recall that for $Z \sim \text{Binomial}(N, \tau)$ we have

$$Z \overset{approx}{\sim} \mathcal{N}\left(N\tau, N\tau\left(1-\tau\right)\right)$$

for N large enough and τ not too close to either zero or one.

Standardizing we get

$$\frac{Z - N\tau}{\sqrt{N\tau\left(1-\tau\right)}} \overset{approx}{\sim} \mathcal{N}\left(0, 1\right).$$

Choosing j and k

Let $j = \lfloor N\tau - l \rfloor$ and $k = \lceil N\tau + l \rceil$ for some l (a "centered" interval).

We want to choose l (and hence j and k) such that

$$\Pr(j \leq Z \leq k) = 1 - \alpha.$$

We can choose an l for which this is approximately true, by cleverly using the connection between the coverage of the interval $\left[Y_{(j)}, Y_{(k)}\right]$, the binomial distribution, and its normal approximation...

Choosing j and k

$$
\begin{aligned}
\Pr\left(j \leq Z \leq k\right) &= \Pr\left(\frac{j - N\tau}{\sqrt{N\tau\left(1-\tau\right)}} \leq \frac{Z - N\tau}{\sqrt{N\tau\left(1-\tau\right)}} \leq \frac{k - N\tau}{\sqrt{N\tau\left(1-\tau\right)}}\right) \\
&\approx \Pr\left(\frac{-l}{\sqrt{N\tau\left(1-\tau\right)}} \leq \frac{Z - N\tau}{\sqrt{N\tau\left(1-\tau\right)}} \leq \frac{l}{\sqrt{N\tau\left(1-\tau\right)}}\right) \\
&\approx \Phi\left(\frac{l}{\sqrt{N\tau\left(1-\tau\right)}}\right) - \Phi\left(\frac{-l}{\sqrt{N\tau\left(1-\tau\right)}}\right) \\
&= 2\Phi\left(\frac{l}{\sqrt{N\tau\left(1-\tau\right)}}\right) - 1 \\
&= 1 - \alpha.
\end{aligned}
$$

Choosing j and k

Let $z^{1-\alpha/2} = \Phi^{-1}\left(1 - \frac{\alpha}{2}\right)$ be the $1 - \frac{\alpha}{2}$ quantile of a standard normal random variable.

With $\alpha = 0.05$ we have $z^{1-\alpha/2} = \Phi^{-1}\left(1 - \frac{\alpha}{2}\right) \approx 1.96$.

Solving we get

$$l = \sqrt{N\tau\,(1-\tau)} \cdot z^{1-\alpha/2}.$$

So if we choose $j = \lfloor N\tau - l \rfloor$ and $k = \lceil N\tau + l \rceil$ with l as above we will have

$$\Pr\left(Y_{(j)} \leq \theta_\tau \leq Y_{(k)}\right) \approx 1 - \alpha.$$

Recipe Recap: Estimation

1. Sort the random sample $Y_1, \ldots, Y_N$ to get the order statistics $Y_{(1)}, \ldots, Y_{(N)}$.

2. Find l such that

$$\frac{l}{N+1} \leq \tau < \frac{l+1}{N+1}$$

and compute

$$\widehat{\theta}_\tau = \frac{Y_{(l)} + Y_{(l+1)}}{2}.$$

Recipe Recap: Confidence Interval

1. Find l such that

$$l = \sqrt{N\tau\,(1-\tau)} \cdot z^{1-\alpha/2}.$$

2. For $j = \lfloor N\tau - l \rfloor$ and $k = \lceil N\tau + l \rceil$, construct the order statistic confidence interval

$$\left[Y_{(j)}, Y_{(k)}\right].$$

Standard Error Estimation

Although it is hard to show, it turns out that

$$\sqrt{N}\left(\widehat{\theta}_\tau - \theta_\tau\right) \xrightarrow{D} \mathcal{N}\left(0, \sigma^2\right)$$

for some σ.

If we knew σ, then we could construct a $1-\alpha$ confidence interval in the normal way:

$$\widehat{\theta}_\tau \pm \frac{\sigma}{\sqrt{N}} \cdot z^{1-\alpha/2}.$$

The *length* of this interval is

$$2 \cdot \frac{\sigma}{\sqrt{N}} \cdot z^{1-\alpha/2}.$$

Standard Error Estimation (continue)

Our order statistic interval is also "centered" around $\widehat{\theta}_T$.

If N is large the length of the two intervals should be about the same:

$$Y_{(k)} - Y_{(j)} \approx 2 \cdot \frac{\sigma}{\sqrt{N}} \cdot z^{1-\alpha/2}.$$

Solving for σ^2 we get

$$\widehat{\sigma}^2 = \frac{N \cdot \left(Y_{(k)} - Y_{(j)}\right)^2}{4 \cdot \left(z^{1-\alpha/2}\right)^2}.$$

Wrapping Up

1. We have defined the τth population quantile, θ_τ.

2. We have introduced a consistent estimator under random sampling.

3. We have presented an order-statistic-based confidence interval which covers θ_τ with an ex ante probability of (about) $1 - \alpha$.

4. We have shown how to use this interval estimate to construct a standard error estimate.

Wrapping Up

Next time we will put all these results together to construct a quantile regression estimator.

We will use this estimator to explore the relationship between education and earnings in Brazil.